Essays

|!|BCEEC0090000001|!|

Describe how lymph is formed and how it can be returned to the blood circulation.

(12 marks)

##

Formation of lymph (Max. 6)

- At the arterial end of the capillary bed, the hydrostatic pressure of the blood is higher than the tissue fluid. (1)
- Some components of the plasma (e.g. water, minerals, glucose and amino acids) (1) are forced out through the differentially permeable capillary walls to become tissue fluid. (1)
- Plasma proteins, red blood cells and blood platelets and remain in the blood due to their large sizes. (1)
- At the venous end of the capillary bed, water moves into the capillary bed by osmosis. (1)
- Excess tissue fluid will enter the lymph capillaries to form lymph (1) due to the higher hydrostatic pressure in the body tissues. (1)

Return of lymph to the heart (Max. 3)

- The contraction of skeletal muscles adjacent to the lymph vessels pushes the lymph forward. (1)
- The presence of valves in lymph vessels prevents the backflow of lymph. (1)
- During inhalation, the decrease in pressure of the thoracic cavity helps draw the lymph towards the heart. (1)
- The large lymph vessels join two large veins at the base of the neck, where the lymph returns to the blood circulation. (1)

Communication (Max. 3)

(12 marks)

##

Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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Compare arteries, veins and capillaries in terms of their features and functions.

(12 marks)

##

Features (Max. 4)

- Arteries have thicker wall to withstand the high blood pressure. (1)
- Veins have thinner wall. (1)
- Capillaries have the thinnest, one-cell thick wall which shortens the diffusion distance for the exchange of materials. (1)
- Arteries possess no valves except at the base of the pulmonary artery and aorta inside the heart. (1)
- Veins possess valves at intervals to prevent the backflow of blood. (1)
- Capillaries have no valves. (1)
- Arteries have smaller lumen. (1)
- Veins have larger lumen to reduce the resistance of blood flow. (1)
- Capillaries have the smallest lumen which is about the same diameter as a red blood cell. This reduces the diffusion distance of oxygen from red blood cells to the body cells. (1)

Functions (Max. 5)

- Arteries carry blood away from the heart. (1)
- Veins return blood towards the heart. (1)
- Capillaries carry blood from arteries to veins in the tissues. (1)
- Arteries (except pulmonary arteries) carry oxygenated blood and nutrients to different body parts. (1)
- Veins (except pulmonary vein) carry deoxygenated blood and metabolic wastes away from different body parts. (1)
- Capillaries allow the effective exchange of substances between blood and body cells. (1)

  This is because:
  - they have one-cell thick wall which shortens the diffusion distance; (1)
  - they are highly branched which increases the surface area for diffusion; (1)
  - their walls are differentially permeable which allows only smaller substances to pass through. (1)

Communication (Max. 3)

(12 marks)

##

Skills assessed	Marks
Understanding basic concepts	9
Communication	3

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|!|BCEEC0090000003|!|

What is lymphatic system? What is its significance in the human body? (10 marks)

##

Main components of the lymphatic system (Max. 2)

- The lymphatic system consists of lymph, lymph vessels and lymph nodes. (1)
- Lymph is formed when some tissue fluid enters the blind-ended lymph capillaries. (1)
- Lymph nodes are swellings present at intervals along lymph vessels. (1)

Significance in the human body (Max. 5)

- Lymph acts as a transport medium, responsible for carrying substances, (1)
  for example white blood cells, (1)
  lipids absorbed by lacteals in the villi of the small intestine. (1)
- Lymph vessels and lymph capillaries complete the circulatory system and (1)
  return the tissue fluid back to the blood circulation. (1)
- Lymph nodes filter the lymph and remove foreign materials from the lymph. (1)
  Lymph nodes are packed with a large number of white blood cells which destroy antigens and help in body defence.

Communication (Max. 3)

(10 marks)

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Skills assessed	Marks
Understanding basic concepts	7
Communication	3

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